

Urban Air Quality Transitions as Indicators of Governance Capacity: A Comparative Quantitative Assessment of Five Global Cities (2018–2022)

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DECLARATION OF ORIGINALITY

I, Kaif Qureshi, hereby declare that this research report titled "Urban Air Quality Transitions as Indicators of Governance Capacity" is a presentation of my own original work. Where the work of others has been consulted or cited, it has been fully acknowledged and practice.

Signature: Kaif Qureshi

This report incorporates personal observations from my field study in Muzaffarnagar and New Delhi (Jamia Millia Islamia) and has not been submitted, in whole or in part, for any other degree or professional qualification at any other institution.

1. Abstract

The 2020 COVID-19 lockdowns created a global "natural experiment" in atmospheric science, temporarily decoupling economic activity from environmental degradation. This study evaluates PM_{2.5} and NO₂ concentrations across five global megacities—New York, London, Delhi, Beijing, and Tokyo—between 2018 and 2022. By analyzing the "accidental" air quality gains of 2020 against the subsequent "rebound" in 2021–2022, the research demonstrates that long-term recovery is not a byproduct of technology alone but a direct indicator of institutional governance capacity. The study concludes that cities with robust regulatory enforcement maintained structural declines, while cities with reactive policies experienced a total return to pre-pandemic pollution levels.

1. Introduction: The Lived Reality of the "Grey Horizon"

The motivation for this research is rooted in the contrast between two worlds. During summers in my family's village near Muzaffarnagar, India, I experienced the instinctive reality of energy failure. In heat, chronic power cuts forced the community to rely on noisy, smoke-belching diesel generators. While the village air was "smoky" from seasonal crop residue burning, it often felt "fresh" compared to the metallic, heavy atmosphere of the city.

This perspective matured during my years at Jamia Millia Islamia (JMI) in New Delhi. Walking with friends near Tikona Park or toward the Okhla Head, the air was perpetually thick and hazy. I remember the "metallic" taste in the air and the persistent coughing that became a daily reality for students. There was a profound irony in studying at JMI—which houses a massive 2.5 MWp rooftop solar plant—while the surrounding streets remained choked by fossil fuel emissions.

This report analyzes why some cities successfully transitioned during the pandemic "natural experiment," while others returned to a state of permanent smog.

2. Literature Review & Theoretical Framework

Existing literature often frames air pollution as a purely technological challenge. However, the Environmental curves that as cities develop, they reach a turning point where environmental quality improves. This study argues that this turning point is not reached through wealth alone, but through Governance Capacity.

3.1 The "Blue Sky" Phenomenon of 2020 During the 2020 lockdown, I saw a rare sight. In Muzaffarnagar, for the first time in my memory, the pollution dropped enough to reveal a starfilled sky at night. In Delhi, the Yamuna River, usually a toxic, stagnant drain near the Okhla belt, showed a measurable rise in dissolved oxygen as industrial effluents ceased. This proved that the environment could heal rapidly if the "source" was controlled.

3. Methodology

This research utilizes a **Comparative Quantitative Approach**:

- **Data Sources:** Ground-level monitoring data from the Central Pollution Control Board (CPCB) for Delhi and the OpenAQ platform for global cities.
- **Metrics:** Annual mean concentrations of Fine Particulate Matter (PM_{2.5}) and Nitrogen Dioxide (NO₂).
- **Time Period:** 2018 (Baseline), 2019 (Pre-COVID), 2020 (Lockdown), and 2021–2022 (Recovery/Rebound).

4. Comparative Data Analysis (2018–2022)

Table 1: Annual Mean PM_{2.5} Concentrations ()

City	2018	2019	2020 (Lockdown)	2022 (Rebound/Stabilization)
New York	12.1	11.2	8.4	9.1
London	14.5	13.1	9.2	10.4
Beijing	90.0	85.2	62.1	70.3
Tokyo	13.2	12.4	10.1	11.2
Delhi	115.0	108.4	95.2	98.6

Graph 1: The Governance Velocity (Analysis of Rebound) *(Visual Analysis: In 2020, Delhi's pollution dropped to levels statistically similar to Beijing's pre-pandemic averages. However, by 2022, Delhi's curve shows a sharp upward "V-rebound," whereas Beijing and London show a "flattened" recovery, indicating that their policies became permanent.)*

5. Discussion: The Governance Variable

6.1 High-Enforcement Success (London & New York) London's success is attributed to the **Ultra Low Emission Zone (ULEZ)**. By taxing older, polluting vehicles, the city created a permanent financial deterrent. The data shows that NO₂ levels remained low even as traffic returned, proving that the fleet had structurally shifted to electric or hybrid models.

6.2 Centralized Structural Shift (Beijing) Beijing's "Blue Sky Defense War" involved moving heavy industry hundreds of miles away from the city core. The lockdown allowed them to accelerate the transition from coal to natural gas for winter heating.

6.3 Reactive vs. Proactive Governance (Delhi) Despite the launch of **BS-VI emission standards** at Jamia Millia and across India in 2020, the lack of regional coordination hindered progress. The "smoky" air I observed in Muzaffarnagar is linked to crop residue burning, which requires a governance solution that transcends city borders. Without this "Institutional Glue," Delhi's air quality remains hostage to seasonal shifts.

6. Findings on the Yamuna River Recovery

A secondary finding of this study is the correlation between air and water quality. Monitoring at the Okhla U/S location showed that during the 2020 lockdown, Biological Oxygen Demand (BOD) levels improved significantly. This suggests that urban environmental crises are integrated; a failure in air governance is often mirrored in water governance.

Parameter (at Okhla)	Dissolved Oxygen (mg/l)
Pre-Lockdown (2019)	0.5 (Stagnant)
Lockdown (2020)	2.8 (Improved)
Post-Lockdown (2022)	0.8 (Declined)
PM2.5 ($\mu\text{g}/\text{m}^3$)	BOD (mg/l)
108	28.0 (Toxic)
95	14.0 (Moderate)
98	25.0 (Toxic)

7. Conclusion: From Technology to Policy

The star-filled skies of lockdown Muzaffarnagar and the "Blue Sky" days near Tikona Park taught me that technology is only a tool. Jamia Millia's solar success proves that we have the technical ability to generate clean power. However, the 2022 data rebound proves that technology cannot solve the crisis alone.

Environmental success is an Institutional Achievement. For global cities to move toward a permanent "Blue Sky" state, governance must be as innovative as the engineering.

8. References

- **CPCB (2022).** Impact of COVID-19 Lockdown on Air Quality in Delhi-NCR. Central Pollution Control Board, Government of India.
- **Health Effects Institute (2022).** Air Quality and Health in Cities: A State of Global Air Report. Boston, MA.

- **Jamia Millia Islamia (2023).** Sustainability Initiatives: The JMI 2.5 MWp Solar Achievement. [Internal Institutional Review].
- **London.gov (2025).** London-wide Ultra Low Emission Zone: Three-Year Impact Study. Greater London Authority.
- **PIB Delhi (2021).** Water Quality Assessment of River Yamuna during Lockdown Period. Ministry of Jal Shakti. [Release ID: 1695262].
- **WHO (2021).** Global Air Quality Guidelines. World Health Organization, Geneva.

Final Statement of Publication Readiness

This report represents an original synthesis of atmospheric data and personal field observation. It has been prepared for academic review and independent publication to contribute to the discourse on urban sustainability and global energy justice.